Scalability & Future Plan

Date	1 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application runs only on a local Flask server.	Users cannot access it online.	High
2	The model uses only four socio-economic indicators.	Prediction accuracy may be limited.	Medium
3	No user authentication or database support.	User data and prediction history cannot be stored.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only local users.	Deploy on a cloud platform (Render, AWS, or Azure) to support multiple users.
2	Data Storage	Uses a static CSV dataset.	Integrate a database such as MySQL or PostgreSQL for storing datasets and prediction history.
3	Performance	Suitable for a small dataset.	Optimize the model and use caching to improve response time for larger datasets.
4	Security	Basic Flask application without authentication.	Add user login, HTTPS, and secure input validation to improve security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy the application on a cloud platform with online access.	2–3 Months	Enables users to access the application from anywhere.
Phase 3	Enhance the prediction model by including additional socio-economic indicators and advanced machine learning algorithms.	4–6 Months	Improves prediction accuracy and reliability.
Phase 4	Add user authentication, database integration, dashboards, and prediction history.	6–12 Months	Provides a more secure, scalable, and feature-rich application.